

Structure Generation

known crystal $\xrightarrow{\text{ICSD matrix}}$ new crystal

Machine Learning

model(features) = targets

Ab-initio Simulations

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}(\mathbf{r}) \right] \varphi_i(\mathbf{r}) = \varepsilon_i \varphi_i(\mathbf{r})$$

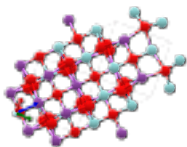
Property and Literature Screening

$$E_{\text{above hull}} < 0.1\text{eV}, E_{\text{gap}} > 0.5\text{eV}$$

Experimental Synthesis and Characterization

reactant 1 + reactant 2 $\rightarrow$ products

mp-756175
 $\text{Zr}_2\text{Bi}_2\text{O}_7$ Fd-3m



mp-1225854:W $\rightarrow$ Te
 CsTaTeO_6 Imma

